Answer **all** the questions in the spaces provided

1 A construction worker on the roof of a hemispherical dome releases a wrench at the highest point A with negligible speed as shown in Fig 1.1. The radius *R* of the dome is 30.0 m. The surface of the dome's roof is smooth. At a certain point B, the wrench just loses contact with the surface of the dome and falls with a projectile motion through the air, and finally hits the ground at point C.

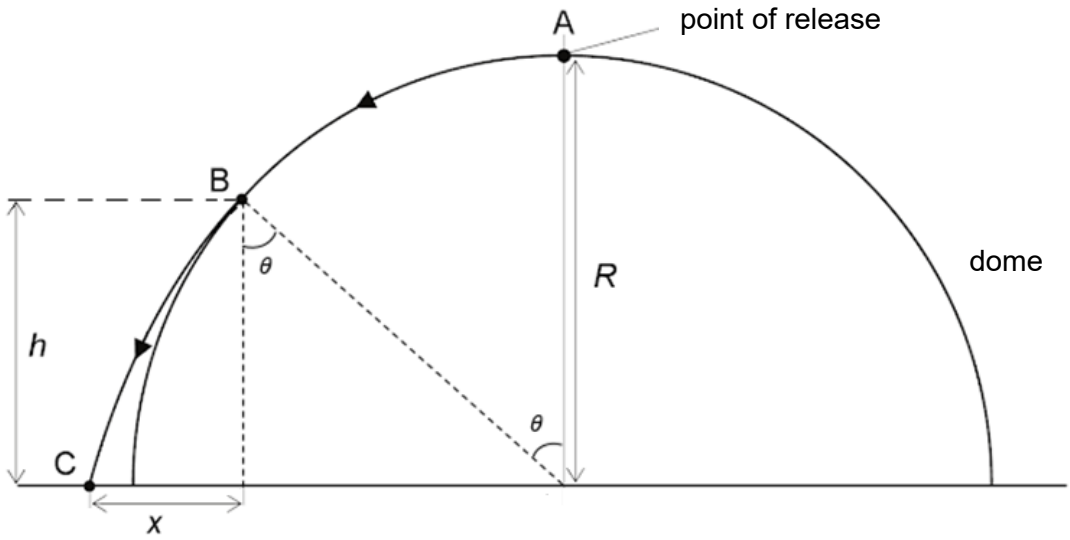

Fig 1.1

(a) **(i)** Explain why the centripetal acceleration of the wrench increases as it slides from A to B.

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……

……

…… [2]

(ii) State the magnitude of the normal contact force on the wrench at point B.

normal contact force = ……..N [1]

(b) By considering the forces contributing to the centripetal force on the wrench at point B, show that the vertical distance between B and C, h is

$$h = \frac{v^2}{g}$$

where v is the linear velocity of the wrench at point B and
g is the acceleration due to gravity.

[3]

(c) **(i)** At the point of losing contact with the surface of the dome, h is related to R by $h = \frac{2R}{3}$.

Use this relation and the relation in **(b)** to calculate for the wrench at the point of losing,

1. its speed and
2. its direction of motion with respect to the horizontal.

speed = …….. m s^{-1}

direction from the horizontal = …….° [2]

(ii) Using the equations of motion, determine the time taken for the wrench to fall from point B to C. Air resistance is assumed to be negligible.

time taken = …….. s [3]

(iii) Hence determine x, the horizontal distance between points B and C.

horizontal distance, x = ……. m [2]